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Native oxidation resistance of $Al_{20}Nb_{30}Ta_{10}Ti_{30}Zr_{10}$ refractory complex concentrated alloy (RCCA)

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ABSTRACT

The native oxidation resistance of an $Al_{20}Nb_{30}Ta_{10}Ti_{30}Zr_{10}$ (at.%) refractory complex concentrated alloy (RCCA) was evaluated at 1000 °C in air. Oxidation tests revealed that the alloy exhibits linear oxidation kinetics out to 4 h of oxidation, with an adherence to parabolic, diffusion-controlled oxidation from 4 to 100 h. This transition was ultimately attributed to the formation of an outer oxide consisting primarily of TiO_2 and $Ti_7Al_2O_{15}$ layers. The 100 h oxide product contained multiple internal layers, but the overall oxidation kinetics was proposed to be controlled by the outer composite oxide layer. While this layer was morphologically continuous, it consisted of a non-continuous mixture of $NbAlO_4$, ZrO_2, $Ti_7Al_2O_{15}$, and $Ta_2Nb_4O_{15}$. The observed oxide scales are discussed relative to the oxidation behaviors of less-complex alloys.

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1. Introduction

Emerging technologies in the areas of advanced propulsion and hypersonics require metallic solutions that exceed the capability of those characteristic of legacy turbine engines. Conditions in these applications push the temperature limits of current material solutions, and in many cases, integrate aggressive stimuli such as shock in combination with a corrosive and/or oxidizing environment. In the realm of metallic solutions, refractory alloys are the most appropriate class of materials due to their inherent high temperature strength and performance [1]. However, it is well known that commercial refractory alloys contain high concentrations of refractory elements (e.g. Nb, Ta, Mo, W) making them susceptible to catastrophic oxidation at high temperatures and therefore prime-reliant on coatings for most applications [1–4]. Refractory complex concentrated alloys (RCCAs) represent a relatively new class of materials that have high concentrations of both common refractory elements (e.g. Nb, Mo, Ta, …) and lower melting point constituents (e.g. Ti, Cr, Al, …) [5–7]. This design concept challenges the conventional approach of utilizing a primary refractory element base (e.g. Nb, Ta) with dilute alloying additions to tailor performance. RCCAs have been reported to exhibit unique combinations of properties, including room temperature ductility, high temperature strength and reduced oxidation kinetics [7–18]. This balance of material characteristics, especially increased native oxidation resistance, would significantly expand the design window for future technologies.

Considering the vast number of RCCA publications that have recently emerged, only a limited number have explored the oxidation behaviors of RCCAs [7–16]. Senkov et al. [7] recently summarized the RCCA oxidation reports in the open literature as part of a larger review. It was reported that as of January 2018, only 20 of the 151 reported RCCAs had been investigated in oxidizing conditions. Reported tests were conducted on a variety of different compositions with various alloy starting conditions (e.g. as-cast, annealed, mechanically-alloyed). All reported oxidation tests were conducted between 700 and 1300 °C, with most tests occurring at 1000, 1100, and 1300 °C. It was concluded that the reported oxide products and corresponding mass data suggest that RCCAs exhibit a wide range of oxidation behaviors, but recommended that more systematic studies need to be done in order to develop a fundamental understanding of the oxidation mechanisms. It was also reported that no studies have systematically studied the influence of microstructure on the resulting oxidation behaviors.

The most oxidation resistant RCCAs have been reported in the works of Gorr et al. [11–13] who investigated AlCrMoTiTa, AlCrMoTiW, and AlCrMoTiNb(Si) alloys. It was reported that the compositional design strategy was to combine common refractory

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elements (e.g. Mo, Ta, or W) with oxidation resistant elements (e.g. Al, Cr, Ti) to promote an alloy with adequate strength and environmental resistance. The AlCrMoTiW alloy had the highest mass gain, while the AlCrMoTiTa alloy had the lowest mass gain (on the order of Cr_2O_3-formers) and exhibited parabolic oxidation kinetics at 1000 °C and 1100 °C up to 40 h. The protection of the AlCrMoTiTa alloy was attributed to the formation of a continuous Al_2O_3 scale that formed below an outer scale of TiO_2. The AlCrMoTiNb alloy exhibited moderate oxidation resistance (<0.16 mg/cm^2 per hour at 1100 °C), but did not adhere to parabolic oxidation kinetics at 1000 °C and 1100 °C. However, moderate improvements to the oxidation kinetics of the AlCrMoTiNb alloy were facilitated by small additions of Si [12]. With that being said, mechanical testing of the AlCrMoTiNb alloy revealed no room temperature ductility in compression [17]. No mechanical testing of the AlCrMoTiW and AlCrMoTiTa alloys has been reported.

Similarly, oxidation tests have been conducted on RCCAs by various other researchers [8–10,14–16]. Butler et al. [8] systematically investigated the oxidation behaviors of NbTiZrV and NbTiZrCr. It was reported that the substitution of Cr in place of V facilitated drastic changes in the alloy microstructures. This ultimately caused large differences in the oxidation kinetics, which were an order of magnitude larger than the Gorr alloys [11–13]. The NbTiZrV alloy exhibited linear oxidation kinetics and was completed oxidized after ~8 h of oxidation at 1000 °C. In contrast, the NbTiZrCr alloy displayed an initial linear regime of oxidation followed by parabolic oxidation from ~1.8 h to 100 h. In both cases, the primary mechanism reported was internal oxidation and each alloy formed a series of composite layers consisting of a combination of simple and complex refractory oxides. This behavior is consistent with oxidation tests of other RCCAs [9,10,14–16]. Butler et al. [8] concluded that the oxides were driven primarily by thermodynamic stabilities and could be reasonably predicted using existing thermodynamic data. It was also deduced that the oxide evolution and progression with prolonged oxidation time could be navigated using an inference from simple oxidation experiments of relevant binary systems (e.g. Nb-Ti, Nb-Zr).

Intermediate temperature oxidation (600–1300 °C) of HfNbTaTiZr-based RCCAs has also been explored [14,16]. Sheikh et al. [14] investigated the native oxidation of the ductile $Hf_{0.5}Nb_{0.5}Ta_{0.5}Ti_{1.5}Zr$ RCCA. It was reported that the alloy exhibited powder disintegration, “pesting”, during oxidation experiments between 600 and 1000 °C for 5 h in air. However, no indication of pesting was observed during oxidation experiments at 1100 °C. This behavior was attributed to the combination of internal oxidation due to the lack of protective scale formation, as well as poor mismatch between the oxide scale and metallic substrate. It was proposed that the simultaneous removal of Zr and Hf could eliminate pesting by reducing the overall volume of oxide in the resulting ternary alloy. However, this could have strong implications on strength, since Zr and Hf are well-known solid solution strengtheners in commercial refractory alloys. Conversely, Chang et al. [16] explored the influence of Al additions in Al_xHfNbTaTiZr alloys on pesting during oxidation experiments between 700 and 1300 °C. It was reported that the Al_0 alloy exhibited pesting at oxidation experiments between 700 and 900 °C. However, Al additions of 10% prevented all pesting behavior between 700 and 900 °C. The Al_{10} alloy was still susceptible to unfavorable oxidation behavior above 1100 °C. The rationale for the elimination of pesting was that the increased Al content allowed for the formation of Al-rich protective oxides.

This brief synopsis reveals that the oxidation characteristics of RCCAs are highly complicated, displaying a wide range of oxidation kinetics and corresponding behaviors. This demonstrates the need for further studies to develop a foundational understanding of RCCA oxidation mechanisms. To date, no unified description and/or predictive modeling capabilities have been widely accepted and implemented to describe RCCA oxidation. To help inform these desired competencies, this work presents a thorough investigation of the oxidation behavior of an AlNbTaTiZr-based RCCA, which is an alloy derived by Senkov et al. [18]. The resulting oxide scales and their evolution with prolonged oxidation exposure will be addressed. The oxidation kinetics will be described in terms of the resulting oxide microstructures and discussed relative to less-complex binary and ternary alloys.

2. Experimental procedures

An alloy (AlNbTaTiZr) with nominal composition $Al_{20}Nb_{30}Ta_{10}Ti_{30}Zr_{10}$ (at.%) was fabricated using vacuum arc melting on a water-cooled copper hearth [18]. Residual contaminants (e.g. oxygen and nitrogen) were minimized by the use of a high purity Ti getter. To ensure uniformity, the alloy button was flipped and subsequently re-melted five times and eventually cast into a cigar-shaped ingot (~100 mm × 30 mm × 12 mm). The entire ingot was wrapped in Ta foil and hot isostatically pressed (HIPed) at 1400 °C for 2 h and 207 MPa (30 ksi). Prior to sectioning, the ingot was annealed at 1400 °C for 24 h under high-purity argon and eventually cooled at 10 °C/min.

Electric discharge machining (EDM) was used to excise specimens (~4.5 mm × 4.5 mm × 7.0 mm) near the center of the ingot. Continuous oxidation tests of the heat treated specimens were conducted with a Thermax 700 Thermogravimetric Analyzer (TGA) at 1000 °C in dry air (80 ml/min). Samples were heated (15 °C/min) under flowing dry air, held at temperature for 100 h, and subsequently cooled (10 °C/min). Oxidation tests were also conducted at the same temperature for multiple times (0.5hr, 4hr, and 100hr) in a box furnace under static laboratory air. Prior to all oxidation tests, samples were prepared to a 600 grit finish and ultrasonically cleaned with acetone and isopropanol. Mass gain measurements of the abbreviated oxidation samples were taken using a high sensitivity balance (10^{-6} g).

Specimens for microstructural characterization were hot-mounted and prepared with conventional metallographic techniques up to a 0.05 μm finish. To preserve oxide scales, oxidized specimens were hot-mounted, sectioned in half with respect to the long dimension using a high precision diamond saw, and remounted in cross-section for further analysis. The resulting microstructures were investigated using a combination of X-ray diffraction (XRD), scanning electron microscopy (SEM), energy-dispersive X-ray spectroscopy (EDS), and transmission electron microscopy (S/TEM). XRD scans of bulk samples were captured in plan-view using a Bruker D2 PHASER (Cu-Kα) and a scan range of 20–100 (2Θ). Secondary electron (SE) and backscattered electron (BSE) images of respective cross-sections were captured using an FEI Quanta 600F SEM at 20 kV. Fib lamella, for subsequent TEM analyses, were made using an FEI Nova 600 Dual Beam FIB-SEM. High angle annular dark field (STEM-HAADF) images and selected area electron diffraction patterns (SADPs) were captured on a 200-KeV FEI Talos F200X (S)TEM equipped with a Super-X (Quad) EDS system.

3. Results

3.1. Microstructure of heat treated alloy

The microstructure and mechanical response of the AlNbTaTiZr alloy has been investigated in the work of Senkov et al. [18], but will be revisited here to provide context for oxidation discussion. The alloy exhibited a coarse grained microstructure (order of

millimeters) in the HIPed and annealed condition. Based on XRD analysis, the alloy appeared single phase body centered cubic (BCC; a = 330.6 pm), but closer examination of the grain boundaries via SEM revealed localized enrichment of Zr and Al. The BCC phase was found to be richer in Nb, Ta, and Ti. For the purpose of oxidation in this work, it was assumed that the free surfaces consisted primarily of single phase BCC, with limited grain boundaries present.

3.2. Oxidation behavior

Fig. 1 (a)-(b) shows linear and logarithmic plots of specific mass change (mg/cm^2) versus time (hours), respectively. For comparison, mass data of two other similar NbTiZr-based alloys is also plotted in Fig. 1 (a) [8]. Continuous (TGA) oxidation data for the AlNbTaTiZr alloy is plotted using a solid line, while the abbreviated oxidation data is plotted as discrete points. In addition, trend lines corresponding to linear ($n = 1$) and parabolic ($n = 0.5$) oxide growth regimes following the general oxidation growth law [2,3] are shown in Fig. 1 (b). From the linear plot, it is evident that the AlNbTaTiZr alloy has lower mass gain than the reference alloys and was found to obey parabolic oxidation kinetics (n = 0.5; $k = 1.4 \times 10^{-3}$ mg cm^{-2} s$^{-0.5}$) for the majority of exposure. Inspection of the logarithmic plot reveals that the oxidation kinetics ultimately consists of a linear regime (n = 1) up to ~0.5 h, a transitional mixed-mode behavior between 0.5 and 4 h, followed by parabolic oxidation kinetics (n = 0.5) out to 100 h. This suggests a period of initial transient oxidation, followed by a subtle transition to diffusion controlled oxidation following 4 h. Therefore, the nature of the oxide scales established at 4 h appears to reduce the overall oxidation kinetics of the alloy. While the abbreviated mass measurements are slightly larger than the continuous data, both tend to follow the same trend.

3.3. Oxidized microstructures

BSE images of the oxidized cross sections from the 0.5hr, 4hr, and 100hr specimens are shown in Fig. 2(a)-(c), (d)-(f), and (g)-(i), respectively. In all three conditions, the oxide constituents appeared to form in variations of a three-layered structure, most clearly observed in the 100 h condition, Fig. 2 (h). This combination consisted of a top, composite "*oxide*" layer, an intermediate "*partially oxidized*" zone with both oxides and remnant metal, followed by the "*internal oxidation front*". The only exception to this multi-layered structure was that a precipitate free "denuded" region was found just below the top oxide in the 0.5 h condition, Fig. 2 (a). Higher magnification imaging of each region revealed that all layers consisted of a complex, fine-scale mixture of phases. In addition, the 100 h oxidized sample exhibited a low-Z contrast precipitate phase under the oxide product and throughout the remaining bulk. This particular precipitate phase was not observed during shorter oxidation experiments, but did form in the alloy during a separate 100 h vacuum annealing experiment at 1000 °C. It is assumed this particular precipitate forms under prolonged exposure at 1000 °C and does not likely manifest itself due to the extent of oxidation.

Table 1 shows the relative thicknesses of each layer with time, as well as the total penetration depth of each layer interface. From this data, it is clear that the top oxide grows to the smallest extent, while the secondary "partially oxidized" layer grows to the largest extent. The total extent of oxidation (e.g. location of the internal oxidation front) was measured visually via SEM to be 39 μm, 139 μm, and 582 μm for the 0.5hr, 4hr, and 100hr exposures, respectively.

Bulk area SEM-EDS (at. %) measurements were captured from all regions of the oxidized specimens, as indicated numerically in Fig. 3 and Table 1. It was determined that the precipitate denuded region in the 0.5 h exposure, layer (2) in Fig. 3 (a), displayed nearly the same average composition as the "partially oxidized" region below it (layer (3)). Surprisingly, the average composition of each respective layer was fairly consistent with prolonged oxidation exposure. The ratio of the metal element concentrations of the layers matched well with the ratio of metal elements considering the bulk alloy composition. This suggests that the diffusion of metal cations during the oxidation process is limited, where the oxidation mechanism occurs by the ingress of oxygen into the alloy causing internal oxidation. This is also consistent with the relative oxygen content and how it decreases in each layer progressing deeper into the alloy, Table 1.

3.4. 100 h oxide layers

TEM analysis was conducted on each respective layer of the 100 h oxidized specimen. FIB-lamella were extracted in cross-section, parallel to the free surface, near the center of each layer, as indicated by the boxes in Fig. 4 (a), Fig. 5 (a), and Fig. 6 (a). Fig. 4 (b)-(n) shows a representative STEM-HAADF image, composite and

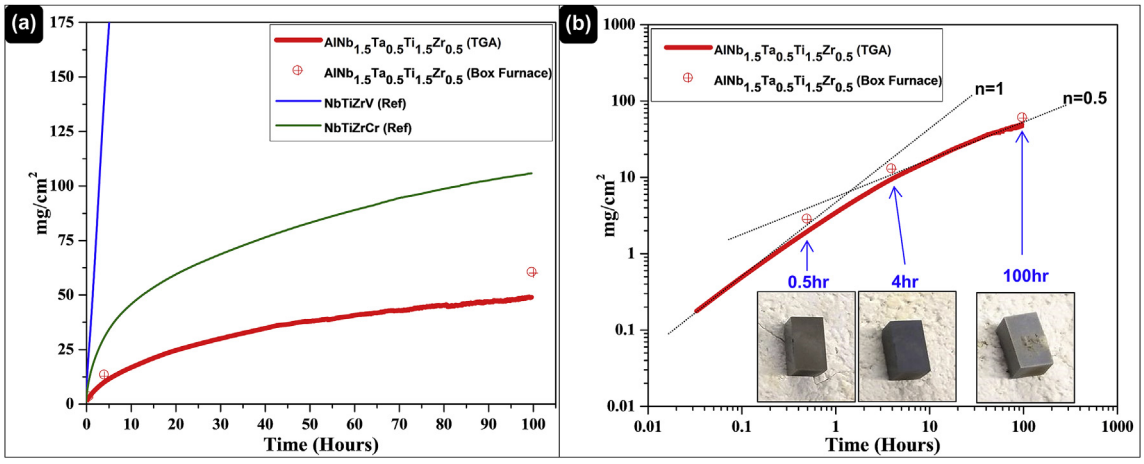


Fig. 1. Specific mass change versus time for the AlNbTaTiZr alloy (a) with continuous lines for TGA data and discrete points for abbreviated oxidation tests; Specific mass change versus time on log/log coordinates with linear (n = 1) and parabolic (n = 0.5) oxide growth trend lines drawn for reference (b). The inset images in (b) are from samples at 0.5hr, 4hr, and 100 h of oxidation, respectively. Reference data in (a) taken from Ref. [8].

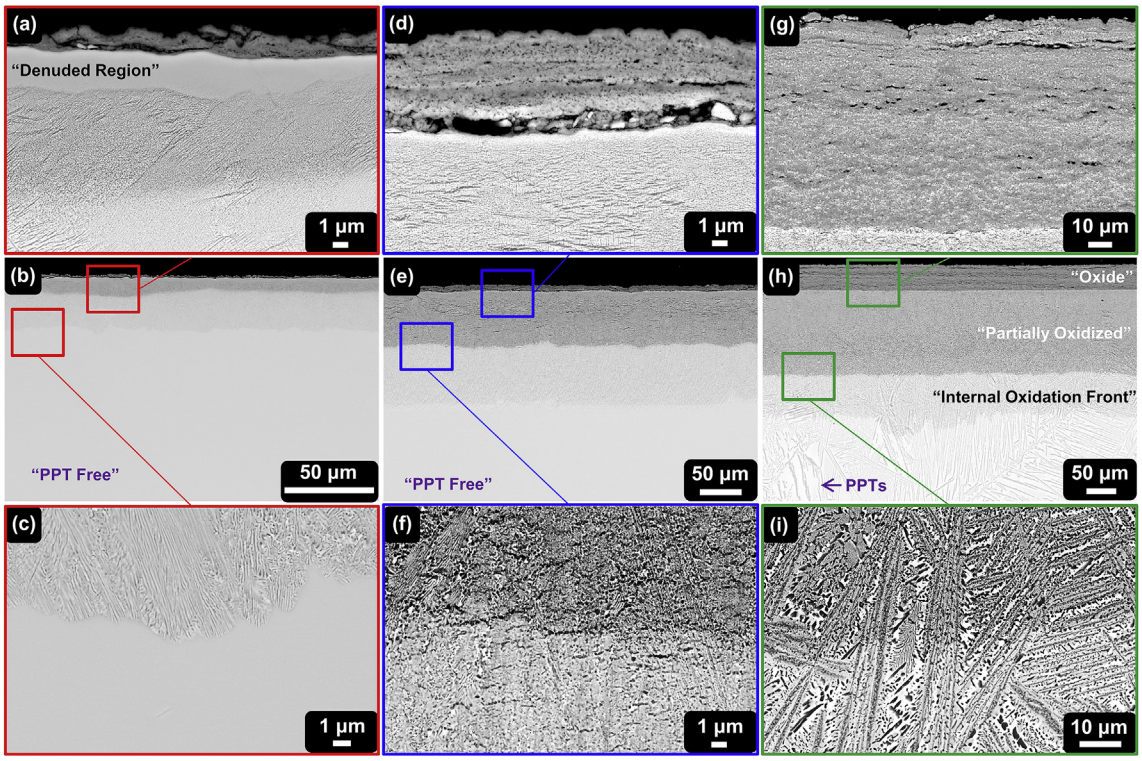


Fig. 2. Cross-sectional BSE images from the 0.5hr, 4hr, and 100hr oxidized alloy, (a)–(c), (d)–(f), and (g)–(i), respectively. Higher magnification images were extracted from inset boxes shown in (b), (e), and (h).

Table 1
SEM-EDS measurements (at. %) captured from the oxidized specimens; thickness and depth measurements of the respective layers in each specimen.

	Region	O	Al	Zr	Nb	Ti	Ta	Thickness (um)	Total Depth (um)
0.5hr	(1)	62.1	6.5	3.0	10.6	13.9	4.0	2	2
	(2)	42.6	11.6	5.4	17.0	16.9	6.4	4	6
	(3)	41.8	11.8	5.6	17.3	17.0	6.5	9	15
	(4)	18.4	15.9	7.8	24.4	24.0	9.5	24	39
4hr	(1)	60.4	8.1	3.8	11.1	12.5	4.2	5	5
	(2)	42.5	11.9	5.9	16.5	17.0	6.2	57	62
	(3)	19.2	16.2	8.3	23.5	24.0	8.8	71	139
100hr	(1)	58.2	8.3	4.2	12.0	12.8	4.5	82	82
	(2)	44.1	11.6	5.8	15.3	17.3	6.0	271	354
	(3)	20.3	15.5	8.9	22.0	24.6	8.6	146	582

elemental EDS maps, and phase specific SADPs from the 100hr top composite oxide scale. The strong degree of elemental partitioning to specific regions, as evident by the contrast in Fig. 4 (b) and chemical color in Fig. 4 (c)-(d), reveals four distinct phases. Based on the oxygen elemental map, Fig. 4 (g), it was determined that each of the four phases were oxides, labeled R1-R4 in Fig. 4 (d). R1 was rich in Nb and Al and exhibited a structure consistent with $NbAlO_4$ (ICDD 00-041-0347, *C*2/*m* (No. 12)). R2 was exclusively rich in Zr and matched a ZrO_2 structure (ICDD 00-036-0420, $P2_1/c$ (No. 14)). R3 was rich in Ti and Al and matched well with a $Ti_7Al_2O_{15}$ structure (04-010-2131, *C*2/*m* (No. 12)). R4 was rich in Ta and Nb and displayed a $Ta_2Nb_4O_{15}$ structure (00-015-0114, *I*4/*m* (No. 87)). A representative SADP for each region is shown in Fig. 4 (k)-(n), respectively. In all cases, the lattice parameters were fairly consistent with the prototype structures. Furthermore, Table 2 shows the average compositions (TEM-EDS) of each region considering only the ratio of metal constituents. Based on the matching prototype structures, the proposed oxides are listed. If it is assumed that Nb and Ta share a strong affinity for the same lattice site, the experimental compositions for the $NbAlO_4$ and $Nb_4Ta_2O_{15}$ phases require Ti to sit on multiple metal lattice sites. With these assumptions, the phase compositions were found to be reasonably consistent with the proposed crystal structures.

TEM analysis of the "partially oxidized" region revealed of total of four distinguishable phases, as evident from the STEM-HAADF image and corresponding composite EDS maps in Fig. 5 (b)-(e). In particular, the region R1 consisted of long chains of high-Z contrast phase, which were rich in Nb and Ta, while correspondingly lean in oxygen, Fig. 5 (e), (f)-(k). SADPs taken from these regions were consistent with a BCC (Im-3m) structure (a = 329.4 pm), which is similar to the structure and starting lattice parameter of the parent alloy, Fig. 5 (l) [18]. Based on an inference to the oxygen elemental map, Fig. 5 (h), it is thought that these areas are remnant metal alloy that has not been oxidized. It is plausible to suggest that these regions offer little resistance to the ingress of diffusing oxygen. Unlike R1, the other intermixed regions (R2, R3, and R4) were all enriched in oxygen and Ti, Zr, or Al, respectively, Fig. 5 (g)-(j). SADPs taken from R2, which was rich in Ti, matched well with a TiO_2

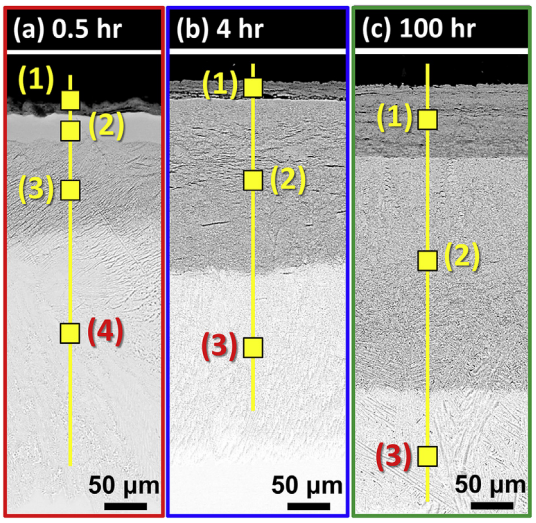


Fig. 3. Cross-sectional BSE images of the alloys oxidized at 0.5hr, 4hr, and 100 h, (a), (b), and (c), respectively. Locations of the composition measurements shown in Table 1 are labeled numerically in each representative image.

structure (ICDD 00-021-1276, $P4_2/mnm$ (No. 136)), Fig. 5 (m). Average compositions considering only metal constituents for all regions are shown in Table 3. Distinguishable SADPs could not be captured from R3 and R4. However, R3 is likely ZrO_2 based on the local Zr and O enrichment. In addition, ZrO_2 was observed and validated in the top composite oxide scale. For R4, one could suggest that due to the high Al content that the oxide may be Al_2O_3. However, based on the composition of R4 shown in Table 3, in conjunction with no clear correlation with the top oxide scale, this region is designated as an Al-rich oxide. Further analysis is required to fully validate the intermediate structures of R3 and R4.

Fig. 6 shows the TEM results from the internal oxidation front. Unlike the previous two layers, the internal oxidation front exhibited a lesser extent of oxidation, as evident from the oxygen elemental map in Fig. 6 (g). It is clear from the STEM-HAADF image and composite EDS maps that this particular region also contains a total of four distinct regions with discrete elemental segregation, Fig. 6 (b)-(d). The average compositions for each region are shown in Table 4. R1 exhibited a complex rod-like geometry and was correspondingly rich in Zr and O. SADPs captured from this region matched a ZrO_2 structure (ICDD 00-036-0420, $P2_1/c$ (No. 14)), Fig. 6 (k). R2, rich in Ti and O, was found to be pervasively distributed about most of the sample area, while some was intermixed with the R1 regions, Fig. 6(c). SAPDs captured from R2 correlated well to a TiO_2 structure (ICDD 00-021-1276, $P4_2/mnm$ (No. 136)), Fig. 6 (l). The remaining regions (R3 and R4) were both lean in oxygen and correspondingly rich in metal elements, Fig. 6 (e)-(j). SADPs from R3, rich in Nb and Ta, matched well with a BCC (Im-3m) structure (a = 329.1 pm), which is similar to the parent alloy structure and the BCC phase observed in the "partially oxidized" region Figs. 5, and 6(m). SADPs captured from R4, rich in Nb and Al, matched well with Ta_2Al (ICDD 01-074-5280, $P4_2/mnm$ (No. 136)), Fig. 6 (n). Considering the composition of R4 in Table 4, it is assumed that Nb shares the Ta lattice site, while Ti sits of both Ta and Al sites.

In order to cross-correlate the proposed oxide structures, XRD was conducted on the free surface (plan-view) of a specimen oxidized for 100 h total, Fig. 7. The experimental spectrum matched well to peaks associated with $Ti_7Al_2O_{15}$, TiO_2, ZrO_2, $Ta_2Nb_4O_{15}$,

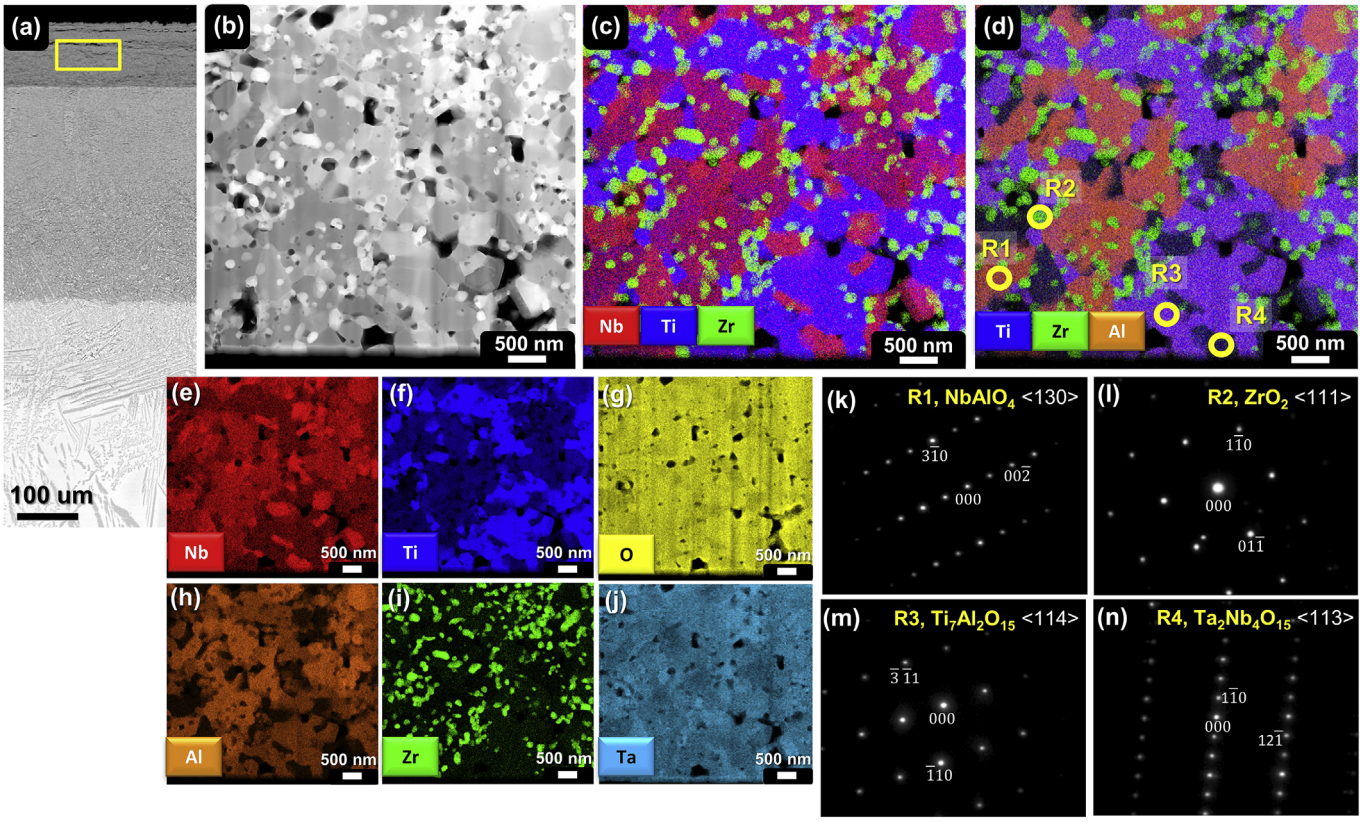


Fig. 4. BSE image of the 100hr oxidized sample (a) with corresponding STEM-HAADF image from a FIB sample extracted from the top layer (b); corresponding composite TEM-EDS maps (c)–(d) of the region shown in (b) and elemental TEM-EDS maps of the same area (e)–(j). (k)–(n) show SADPs captured from the regions labeled R1, R2, R3, and R4 in (d).

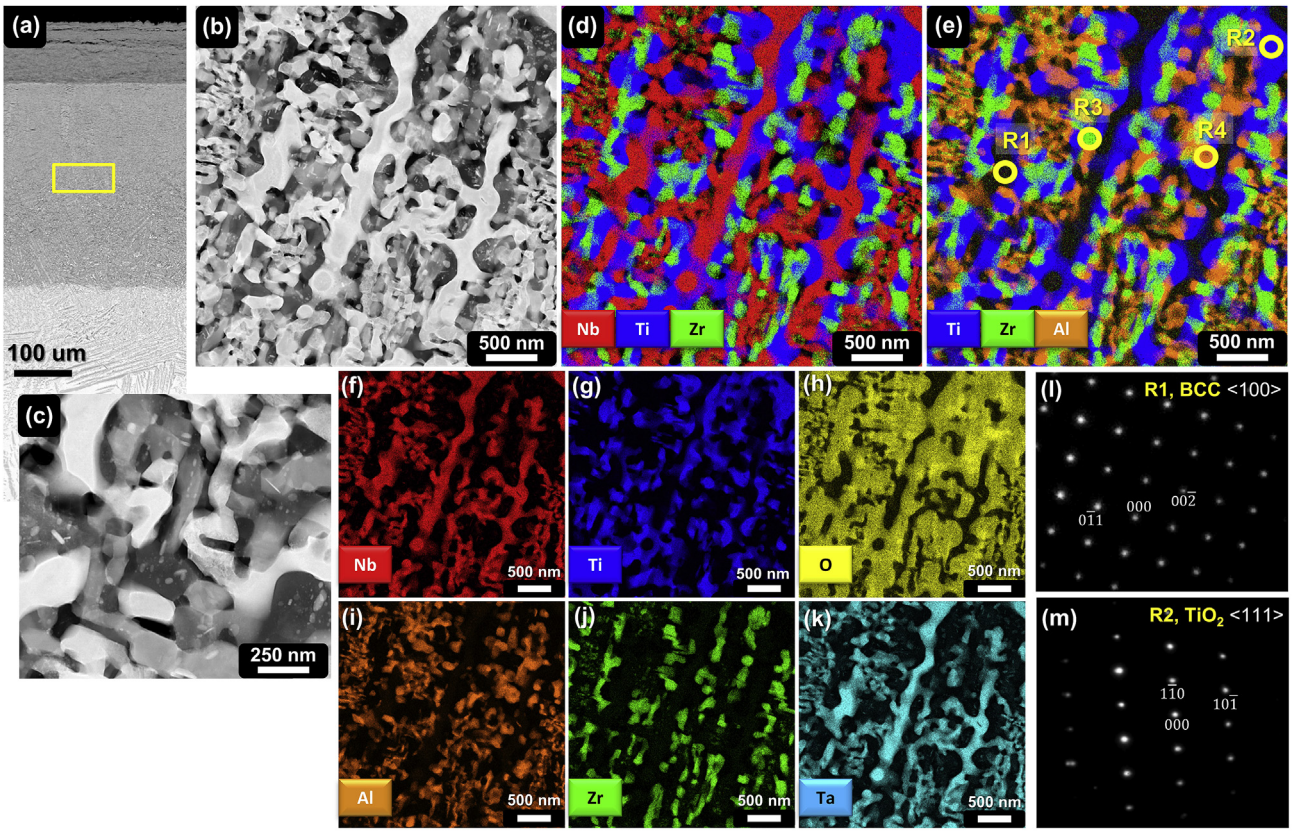


Fig. 5. BSE image of the 100hr oxidized sample (a) with corresponding STEM-HAADF image from a FIB sample extracted from the partially oxidized region (b)–(c); corresponding composite TEM-EDS maps (d)–(e) of the region shown in (b) and elemental TEM-EDS maps of the same area (f)-k). (l)–(m) show SADPs captured from the regions labeled R1 and R2 in (e).

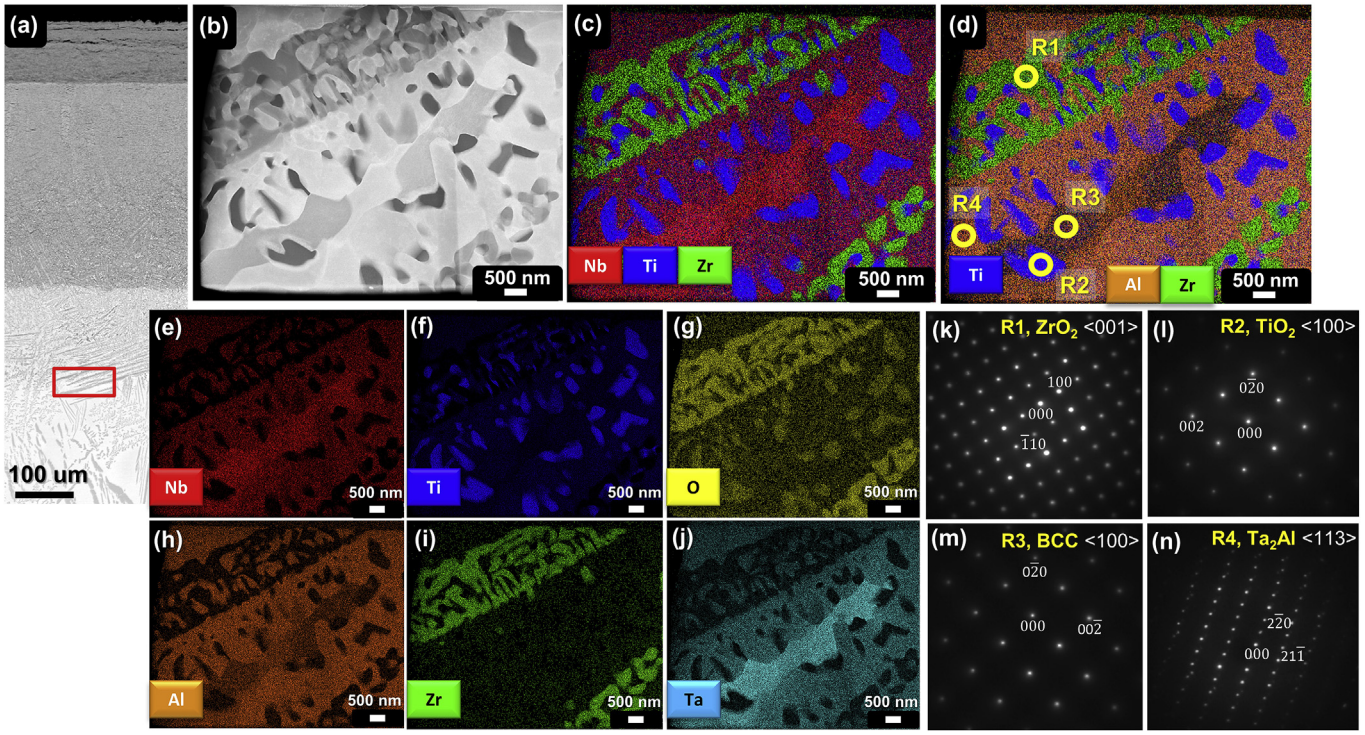


Fig. 6. BSE image of the 100hr oxidized sample (a) with corresponding STEM-HAADF image from a FIB sample extracted from the internal oxidation front (b); corresponding composite TEM-EDS maps (c)–(d) of the region shown in (b) and elemental TEM-EDS maps of the same area (e)–(j). (k)–(n) show SADPs captured from the regions labeled R1, R2, R3, and R4 in (d).

Table 2
TEM-EDS measurements (at. %) captured from the oxides in the top composite oxide scale (100hr). The proposed prototype structures and lattice site preference are also shown.

Region	Nb	Ti	Zr	Al	Ta	Matching Structure	Proposed Oxide
R1	31.9 ± 0.6	19.8 ± 0.6	1.0 ± 0.3	36.7 ± 0.2	10.6 ± 0.7	$NbAlO_4$	$(Nb + Ta + Ti)(Al + Ti)O_4$
R2	18.3 ± 3.4	5.7 ± 2.5	54.9 ± 4.6	3.4 ± 1.3	17.8 ± 3.2	ZrO_2	ZrO_2
R3	16.6 ± 0.9	47.6 ± 0.8	2.4 ± 0.2	21.4 ± 0.7	12.0 ± 0.9	$Ti_7Al_2O_{15}$	$(Nb + Ti + Ta)_7Al_2O_{15}$
R4	50.5 ± 0.8	18.0 ± 1.2	3.3 ± 0.4	5.8 ± 1.0	22.4 ± 1.1	$Nb_4Ta_2O_{15}$	$(Nb + Ti)_4(Ta + Ti)_2O_{15}$

Table 3
TEM-EDS measurements (at. %) captured from the oxides in the "partially oxidized region" (100hr). The proposed prototype structures and lattice site preference are also shown.

Region	Nb	Ti	Zr	Al	Ta	Matching Structure	Proposed Phase
R1	56.9 ± 1.4	4.1 ± 0.5	1.5 ± 0.1	2.1 ± 0.1	35.3 ± 2.0	BCC	BCC
R2	2.3 ± 1.9	92.3 ± 4.2	0.4 ± 0.1	1.6 ± 1.1	3.3 ± 1.1	TiO_2	TiO_2
R3	3.2 ± 0.5	10.3 ± 0.5	75.2 ± 4.5	5.2 ± 2.1	6.3 ± 1.0	–	ZrO_2
R4	10.5 ± 2.8	3.2 ± 0.7	0.8 ± 0.1	79.4 ± 3.2	6.1 ± 0.8	–	Al-Oxide

Table 4
TEM-EDS measurements (at. %) captured from the oxides near the internal oxidation front (100hr). The proposed prototype structures and lattice site preference are also shown.

Region	Nb	Ti	Zr	Al	Ta	Matching Structure	Proposed Phase
R1	2.4 ± 0.5	2.0 ± 0.4	86.4 ± 2.6	3.4 ± 0.6	5.8 ± 0.9	ZrO_2	ZrO_2
R2	3.3 ± 0.7	88.5 ± 1.7	1.3 ± 0.3	3.1 ± 0.9	3.8 ± 0.5	TiO_2	TiO_2
R3	49.6 ± 0.7	8.8 ± 1.1	1.8 ± 0.2	6.4 ± 0.7	33.4 ± 2.0	BCC	BCC
R4	37.0 ± 0.8	15.9 ± 0.4	2.8 ± 0.3	25.9 ± 1.1	18.4 ± 0.3	Ta_2Al	$(Nb + Ta + Ti)_2(Ti + Al)$

$NbAlO_4$, and BCC. Therefore, the peak signatures appear to be consistent with the phases determined from TEM electron diffraction and chemical analysis experiments.

3.5. Abbreviated oxide layers

To better understand the mechanism behind the transition in oxidation kinetics, TEM analyses were conducted on the abbreviated samples. Figs. 8 and 9 show results captured on FIB lamella that were lifted out of the cross-sections in the 0.5hr and 4hr samples, respectively. The intention was to provide an enhanced view of oxide progression from outermost oxide to the "partially internally oxidized" region. In general, the extent of outer oxide scale formation was less pronounced in the 0.5hr compared to the 4hr, Figs. 8 (a)-(b) and 9 (a)-(b), respectively. Considering the elemental maps in Fig. 8 (c)-(h), the 0.5hr outer oxide scale appeared fairly random in distribution and exhibited only small regions of localized segregation. However, both the 0.5hr and 4hr outer oxide scales displayed thin surface enrichment layers consisting primarily of Nb and Al, Figs. 8 (b) and 9 (b), respectively. This infers that some level of small-scale cation diffusion does take place during intermediate oxidation times, but is minimal in extent when considering the average composition of the entire layer. The 4hr outer oxide scale was better-developed and exhibited regions with distinct elemental segregation, Fig. 9 (b)-(h). Surprisingly, the majority of the scale was enriched in a fairly-continuous layer of Ti-rich phase, with randomly distributed Zr-rich particles, and several thin intermediate layers enriched in Al, Fig. 9 (b) and (f). In both conditions, the internal oxidation product appears to be similar, but coarsens with prolonged oxidation time, Figs. 8 (a) and 9 (a).

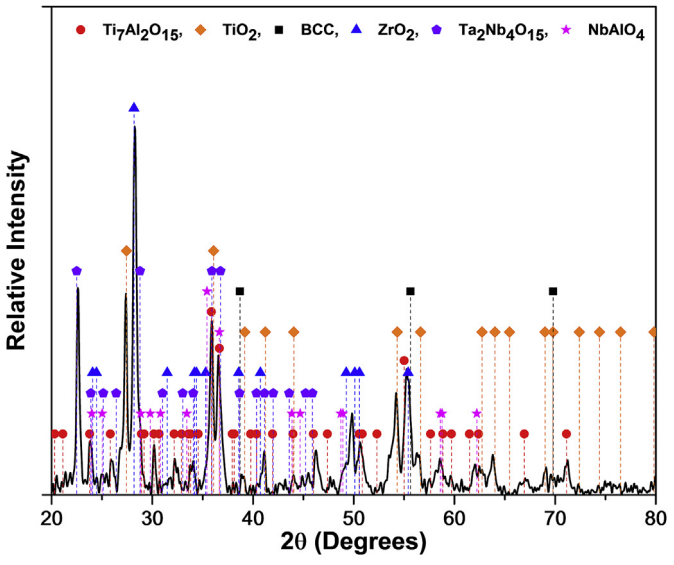


Fig. 7. XRD spectra from the 100hr oxidized specimen. Peaks associated with $Ti_7Al_2O_{15}$, TiO_2, BCC, ZrO_2, $Ta_2Nb_4O_{15}$, and $NbAlO_4$ phases are shown for reference.

Both abbreviated conditions were also explored using XRD, as shown in Fig. 10. For the 0.5hr condition, Fig. 10 (a), the primary signature correlated to BCC, with subtle contributions from ZrO_2, TiO_2, Nb_2O_5, and $Ti_7Al_2O_{15}$. This associates well to the thin and nearly randomly distributed outer oxide scale observed for the 0.5hr condition. Here, the strong presence of BCC originates from the underlying alloy substrate. For the 4hr condition, Fig. 10 (b), the XRD signature displayed strong peaks associated with TiO_2, ZrO_2, and $Ti_7Al_2O_{15}$, with only minor contributions from BCC and Nb_2O_5 present. Due to the distribution and morphology of the Zr-rich phase in Fig. 9 (b) and (g), it is assumed that this phase would offer little resistance to oxygen ingress. Therefore it is plausible to suggest that the reduction in oxidation kinetics after 4 h likely results from the establishment of the TiO_2 and $Ti_7Al_2O_{15}$ layers.

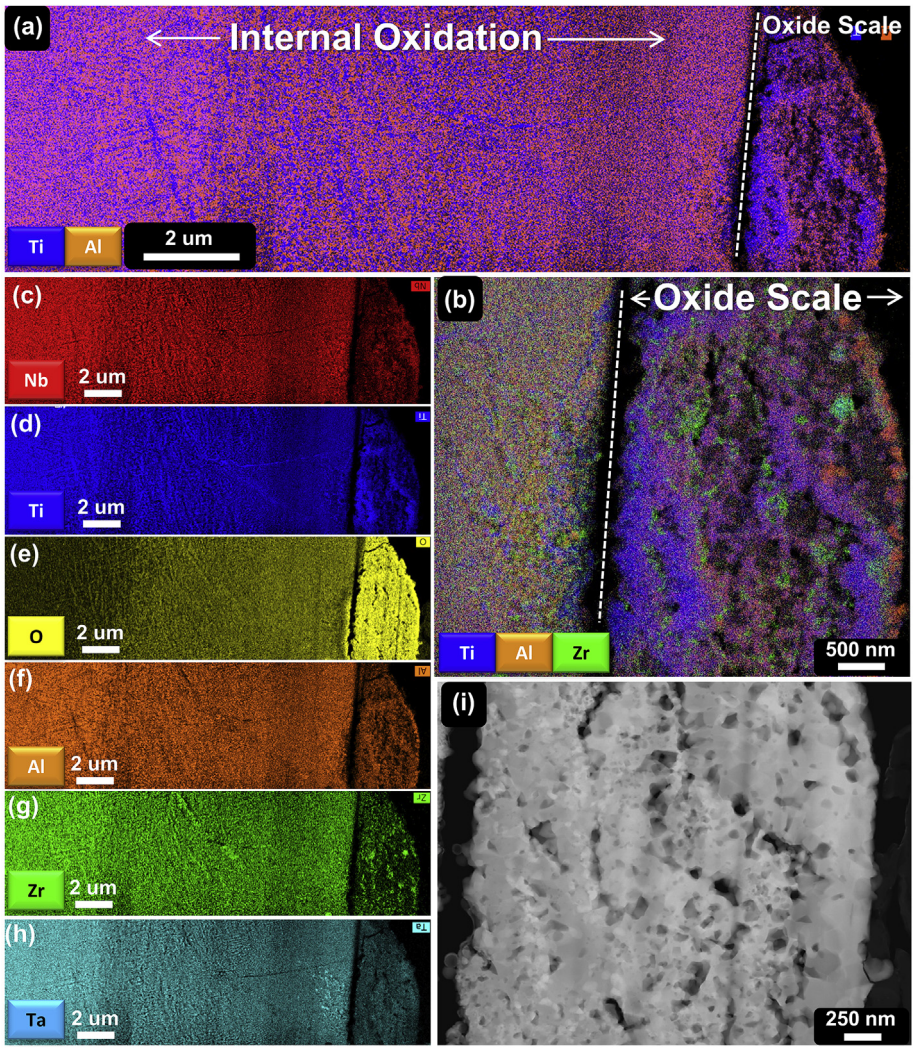


Fig. 8. TEM analysis of 0.5hr oxidized alloy. Composite TEM-EDS maps (a)–(b) of the outer oxide scale; corresponding elemental TEM-EDS maps (c)–(h); and STEM-HAADF image of the outermost scale (i).

4. Discussion

4.1. Oxidation mechanisms

The oxidation behavior of the AlNbTaTiZr RCCA was found to form a variety of complex refractory oxides (e.g. $NbAlO_4$, $Nb_4Ta_2O_{15}$, $Ti_7Al_2O_{15}$), along with simple oxides (e.g. ZrO_2, TiO_2) depending on penetration depth and oxidation time. Internal oxidation was observed to be the primary mode of oxidation. Overall, the oxidation kinetics exhibited a two-stage, linear to parabolic transition after approximately 4 h, Fig. 1. This was attributed to both the morphology and compositional nature of the outer oxide scales when comparing the 0.5 h and 4 h conditions, Figs. 8–10. After 4 h of oxidation, the oxide scale was morphologically better developed and was observed to contain large regions of Ti and Al rich layers that were not evident after 0.5 h of oxidation, Figs. 8(b) and 9(b). These Ti and Al rich regions in the 4 h oxidized condition correlated structurally to TiO_2 and $Ti_7Al_2O_{15}$, Fig. 10(b), indicating that this combination of oxides and morphology contributed to the transition to parabolic oxidation kinetics and reduction in the overall mass gain.

While the types of oxides tended to vary with oxidation duration, it is interesting to note that all oxidized conditions displayed a similar three-layered structure, Fig. 2. These layers consisted of an outer composite oxide layer, an internal "partially oxidized" region with remnant alloy, and the internal oxidation front. To better illustrate the dynamics of each layer, the total depth and the relative layer thickness are plotted in Fig. 11. The trend of each layer depth, Fig. 11(a), is consistent with the change to diffusion-controlled oxidation kinetics after 4 h. In light of the moderate oxidation kinetics, this plot re-emphasizes the fact that the total oxidation interaction was close to 600 μm in extent. The oxygen concentration in either of the inner layers (or potentially in the bulk near the internal oxidation front) may be sufficient to cause substantial embrittlement. The relative thickness of each layer, Fig. 11 (b), reveals that the outer most composite oxide layer grows the slowest throughout the oxidation exposure and is likely the limiting factor reducing the inward oxygen flux and corresponding oxidation kinetics. This outer layer tends to reduce the overall growth rates of the secondary layers, which subsequently appear to offer limited resistance to inward diffusing oxygen.

These results suggest that the outer layer of the AlNbTaTiZr RCCA, once established, facilitates a reduction in the oxidation kinetics and provides moderate mass gain out to 100 h. The outer layer was the only fully oxidized region observed, since the subsequent layers contained remnant metal alloy. Investigation of this

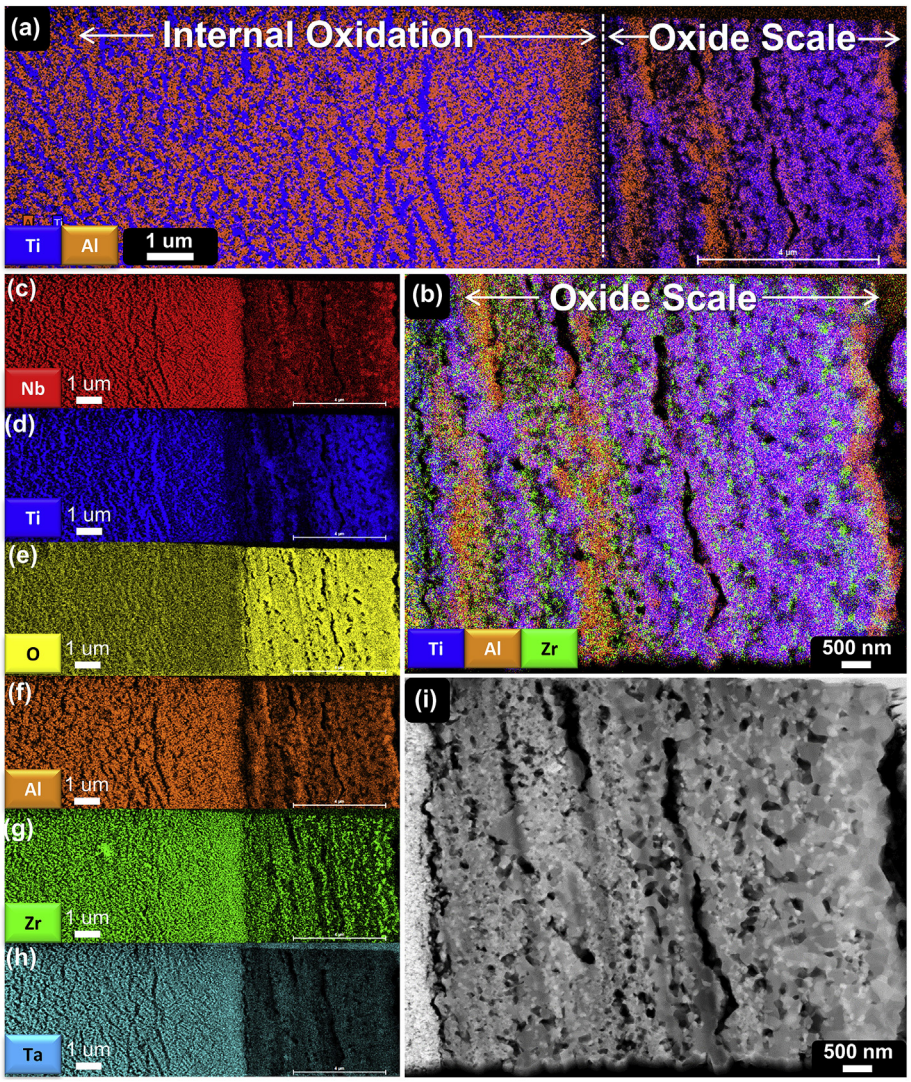


Fig. 9. TEM analysis of 4hr oxidized alloy. Composite TEM-EDS maps (a)–(b) of the outer oxide scale; corresponding elemental TEM-EDS maps (c)–(h); and STEM-HAADF image of the outermost scale (i).

region after 100 h of oxidation revealed a long-term composite oxide layer consisting of a complex mixture of ZrO_2, $NbAlO_4$, $Nb_4Ta_2O_{15}$, and $Ti_7Al_2O_{15}$, Fig. 4. While the outer layer itself was morphologically continuous, none of the individual oxides formed continuous layers.

The implications of this intricate morphology, coupled with the presence of complex oxides, on reducing the oxidation kinetics are not well understood. Butler et al. [8] also observed the formation of discontinuous complex oxide mixtures for NbTiZrV and NbTiZrCr RCCAs. It was reported that the superior oxidation resistance of the NbTiZrCr alloy was due to the formation of rutile-structured $NbCrO_4$ with finely dispersed ZrO_2. In addition, it was mentioned that the oxide of the NbTiZrCr alloy was more dense/compact and exhibited less porosity than the NbTiZrV alloy. Surprisingly, the morphology of the 100hr outer oxide scale that formed on the AlNbTaTiZr alloy was qualitatively comparable in porosity and oxide size to the 100hr scale that was reported for the NbTiZrCr alloy, Fig. 4, [8]. Given that the AlNbTaTiZr alloy exhibits superior oxidation kinetics in comparison to the NbTiZrCr alloy, Fig. 4, and formed a different distribution of oxide products, it is plausible in this case that the oxide species play a dominant role in reducing the oxidation kinetics compared to the complex morphology.

Some inferences to related systems can be made regarding the oxide species that formed in the outer oxide scale. Allouard et al. [19] studied the influence of Ti and Al concentrations on the oxidation behaviors of NbTiAl alloys. It was reported that at intermediate Al concentrations, $NbAlO_4$ was formed and helped facilitate a reduction in the oxidation kinetics compared to alloys that formed primarily Nb_2O_5. Similarly, it was found that small additions of Cr and V to a Nb-40%Ti-42%Al (wt. %) alloy promoted a combination of TiO_2, $TiAl_2O_5$, and Al_2O_3 oxides, which reduced oxidation kinetics as well. Compared to the high oxidation kinetics typically exhibited by the common refractory elements in the AlNbTaTiZr RCCA (e.g. Nb to Nb_2O_5, Ta to Ta_2O_5) it is plausible to expect some level of oxidation resistance from the $NbAlO_4$ and $Ti_7Al_2O_{15}$ oxides observed in this work.

In regards to $Nb_4Ta_2O_{15}$ it is unclear whether the oxide would offer oxidation resistance. Work by Taylor et al. [20] investigated the influence of Ta on the oxidation mechanisms of Nb. It was reported that Ta exhibits a beneficial influence to the oxidation resistance of Nb since it increases the temperature required to form Nb_2O_5 (in light of NbO_z type oxides). This is beneficial since Nb_2O_5 has a low melting point (~1520 °C) and exhibits a large volumetric change during formation. Ta additions also extend the time it takes

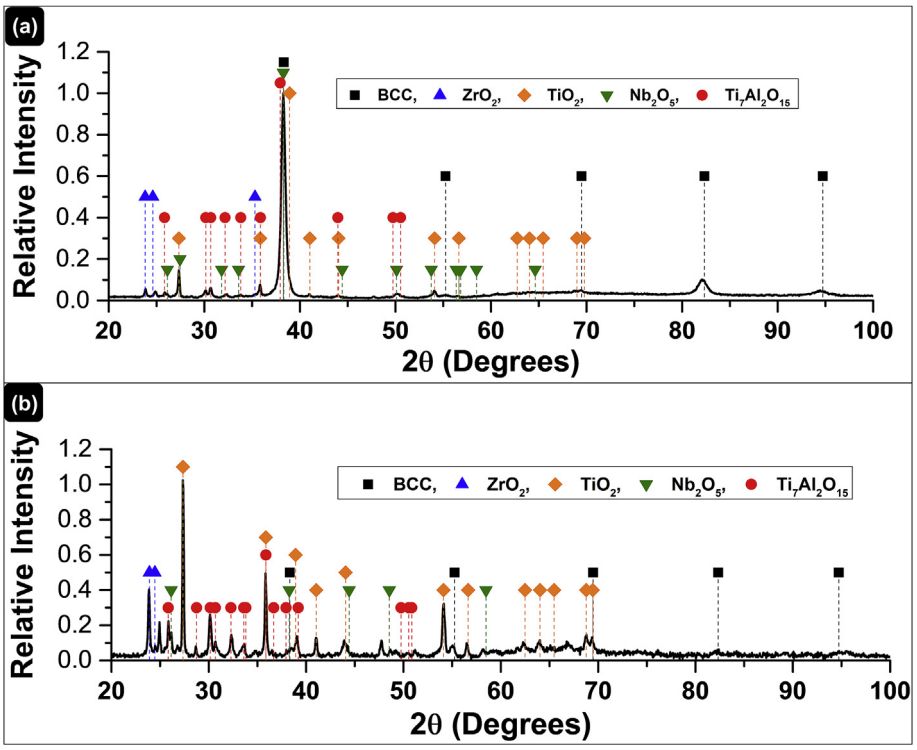


Fig. 10. XRD spectra captured from the 0.5hr and 4hr oxidized conditions, (a) and (b), respectively. Peaks associated with BCC, ZrO_2, TiO_2, Nb_2O_5, and $Ti_7Al_2O_{15}$ phases are listed for reference.

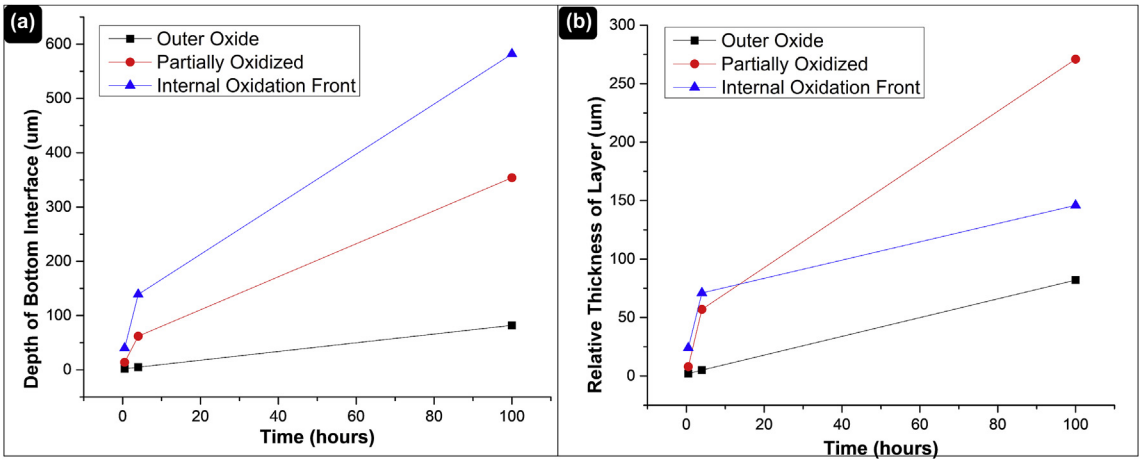


Fig. 11. The measured depth of the bottom interface in each respective layer (a); and relative thickness measurements for each layer (b).

to reach "break-away" oxidation in Nb-Ta, whereby the oxidation kinetics rapidly accelerates material failure. In light of this discussion, $Nb_4Ta_2O_{15}$ may provide an alternative to forming detrimental oxides like Nb_2O_5 and Ta_2O_5. It is not anticipated that ZrO_2 would provide much oxidation resistance, but could serve in some capacity as an initial oxygen getter [8,19].

The strong tendency of internal oxidation in RCCAs could be linked to the high solubility and reactivity of many of the prime elements with oxygen [21]. However, the behavior observed in this and other bodies of work on RCCA oxidation may be biased by the starting microstructures. The AlNbTaTiZr RCCA displayed large grains on the order of millimeters that could have significant implications on the resulting oxidation mechanisms. The observed lack of cation diffusion in this alloy may be due to compositional complexity, but further work investigating the influence of grain boundaries on diffusion and the resulting oxides is required to understand the impact of each of these effects. It is unknown whether smaller grain sizes would promote a beneficial, detrimental, or no influence on the observations in this work. An analogy relevant to this discussion of grain size modification and oxidation mechanisms can be made. Trindade et al. [22] investigated the influence of grain size modification on two low Cr containing steels and three high Cr containing steels. Reducing the grain size in the high Cr content steels facilitated a beneficial result on the oxidation mechanism. It was reported that the high Cr content steels exhibited a strong preference for the selective

oxidation of Cr to form Cr_2O_3, which is highly protective at intermediate temperatures. The integration of more grain boundaries (smaller grain size) produced fast diffusion pathways for Cr that expedited the formation of Cr_2O_3. However, for the low Cr content steels, which do not readily form Cr_2O_3 and prefer the formation of less-protective Fe-oxides, reducing the grain size caused a detrimental effect to the overall oxidation resistance. Without the selective oxidation of Cr, the low alloy steels formed non-protective oxides that did not limit oxygen ingress. This, coupled with the presence of more grain boundary diffusion pathways (smaller grain size), enhanced the inward diffusion of oxygen and the overall extent of internal oxidation. Future work is underway exploring the role of grain boundaries on the oxidation mechanisms in the AlNbTaTiZr RCCA.

4.2. New design paradigms

The concept of RCCAs challenges conventional norm when it comes to alloy design practices. In addition, RCCA oxidation behaviors have created a unique application space that merits further exploration. Generalized oxidation regimes relevant to refractory alloys are shown in Fig. 12. It is not surprising that dilute, uncoated refractory alloys (shown in green) exhibit the highest oxidation kinetics and are the least oxidation resistant solutions. While susceptible in oxidizing environments, these alloys generally display the highest strengths and can have some degree of room temperature ductility (e.g. C-103, TZM). Typical alloying strategies in this regime have emphasized strength over environmental resistance, and the majority of these types of alloys are prime-reliant on coatings [4,23]. In the opposite end of the spectrum (shown in red), refractory alloys that form classically protective scales (e.g. Al_2O_3, Cr_2O_3, and/or SiO_2) inherently exhibit the lowest oxidation kinetics and are the most protective solutions considering environment alone. However, formation of these scales as a long-term, native oxidation product is likely unrealistic, since it requires significant concentrations of Al, Cr, and/or Si, which can ultimately lead to embrittlement of the alloy and loss of toughness [17,24,25]. Therefore, promoting these types of oxides for refractory alloys has been commonly done through coatings [23] and local surface enrichment techniques [26].

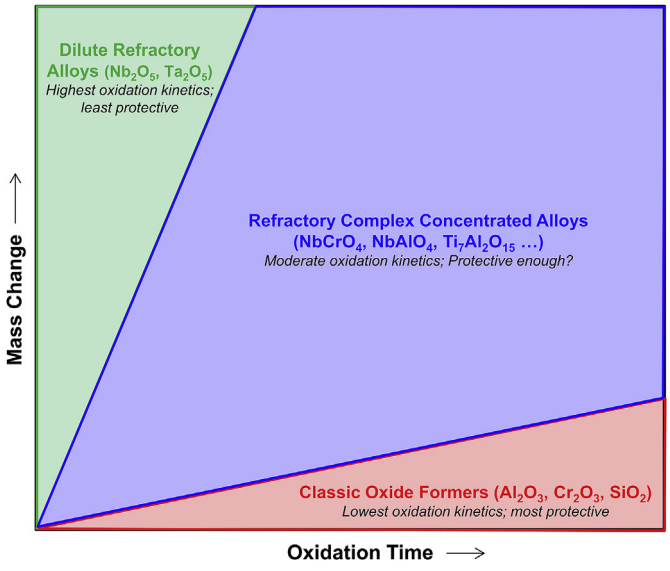


Fig. 12. Generalized oxidation kinetics design scheme, plotting mass change versus oxidation time. Three distinct regions of kinetics are identified.

Little work has explored the regime most relevant to RCCAs, where a combination of moderate strength and moderate oxidation kinetics has been observed [7]. RCCAs could potentially provide extended life in the presence of coating failure or for expendable applications, offer uncoated material solutions when active cooling considerations are design limited. RCCAs demonstrate great potential for new and emerging technologies where high temperature strength alone is not always the primary failure mode. Their compositional complexity offers a broad range of oxide products that are not readily assessable in dilute refractory alloys. Future work exploring the merit of these native oxide products considering relevant environmental stimuli (e.g. temperature $\geq$1200 °C, low or high P_{O2}, shock) is needed.

5. Conclusions

This paper investigated the oxidation behavior (1000 °C/air) of an $Al_{20}Nb_{30}Ta_{10}Ti_{30}Zr_{10}$ (at.%) RCCA. The following key points can be made:

1) Discontinuous and continuous oxidation tests in air revealed that the AlNbTaTiZr alloy exhibited moderate oxidation kinetics out to 100 h (~50 mg/cm^2). The oxidation proceeds initially via linear oxidation kinetics and then adheres to parabolic, diffusion-controlled oxidation kinetics ($n = 0.5$; $k = 1.4 \times 10^{-3}$ mg cm^{-2} s$^{-0.5}$) from ~4 to 100 h. The transition in behavior from linear to parabolic oxidation kinetics was reported to be due to the formation of a dense outer oxide region consisting primarily of TiO_2 and $Ti_7Al_2O_{15}$ layers. The dominant mode of oxidation throughout appears to be internal oxidation.
2) The oxide product consisted primarily of three main zones throughout exposure, namely: an outer composite oxide layer, an internal "partially oxidized" layer, and the internal oxidation front. After 100 h of oxidation, these regions consisted of the following prototype phase structures: *Outer composite oxide*: $NbAlO_4$, ZrO_2, $Ti_7Al_2O_{15}$, and $Ta_2Nb_4O_{15}$; *"Partially oxidized region"*: BCC (metal), TiO_2, ZrO_2, and Al-rich oxide; *Internal oxidation front*: BCC (metal), TiO_2, ZrO_2, and Ta_2Al.
3) The slowest growing layer was observed to be the outer composite oxide, consisting of $NbAlO_4$, ZrO_2, $Ti_7Al_2O_{15}$, and $Ta_2Nb_4O_{15}$. It is proposed that this complicated layer provides an inherent resistance to the inward flux of oxygen, thus slowing the overall oxidation rate. The implication of the complicated morphology of this layer on the oxidation resistance is unknown, but reasonable deductions of the individual protective nature of most of the oxides can be made considering simple systems (e.g. Nb-Ta, Nb-Ti-Al).

Conflicts of interest

"None".

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